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INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES  
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# **Banking Ledger Management with Automated Interest Computation Using Structures, Functions, and Persistent File Storage**

**C – Programming (CSA0201)**

**Under the Guidance of :**

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**Presented By :**

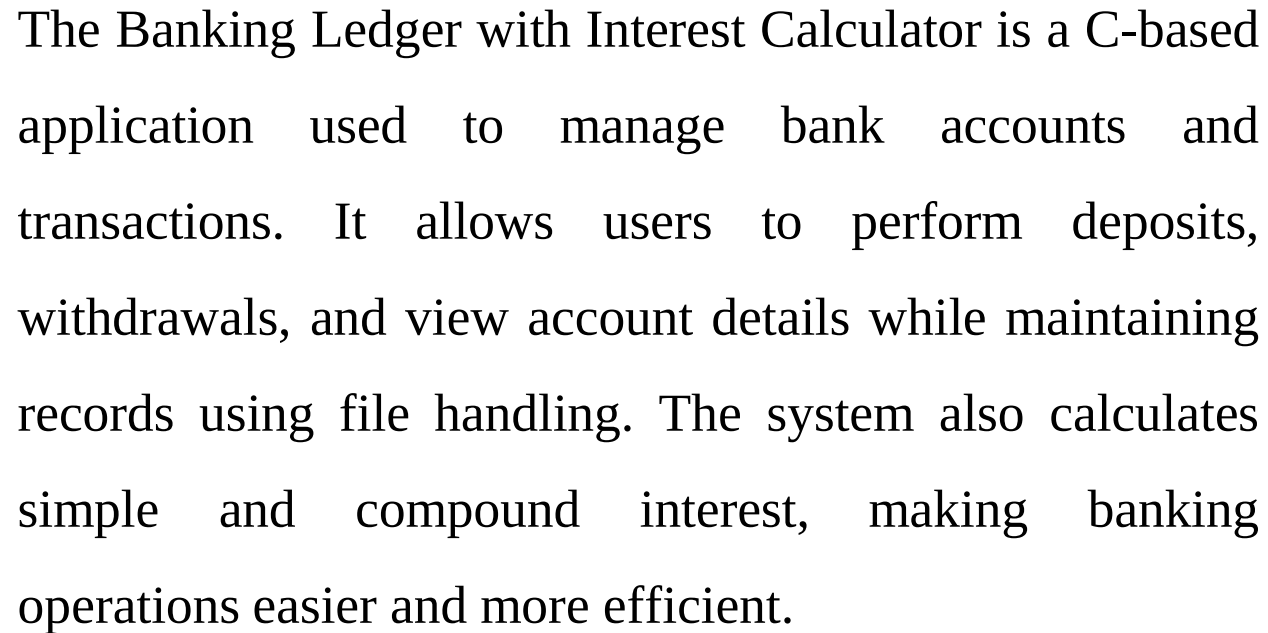
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# Objectives



- To develop a banking ledger system for efficient management of customer accounts.
- To provide facilities for creating, viewing, and updating account information.
- To perform deposit and withdrawal transactions with automatic balance updates.
- To maintain and display a complete transaction history for each account.
- To calculate simple and compound interest accurately based on account details.
- To store account and transaction records permanently using file handling.



# Literature Review

| S. No. | Author(s)                         | Year | Title of the Paper                                                                                                                                 |
|--------|-----------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Megha Garg & Parveen Kumar        | 2024 | Harnessing Digital Technologies for Triple Bottom Line Sustainability in the Banking Industry: A Bibliometric Review                               |
| 2      | Rajesh Kr. Singh et al.           | 2025 | Applications of Artificial Intelligence for Optimizing Banking and Financial Operations: A Systematic Literature Review and Future Research Agenda |
| 3      | J. L. Alonso-Rocha et al.         | 2025 | From Manual to Automated: A State-of-the-Art Review to Examine the Impact of Intelligent Document Processing in Banking Automation                 |
| 4      | Namanyay Goel & Reeta Mishra      | 2025 | The Role of AI in Modern Accounting: Developing an AI Integrated Bank Statement Classifier and Ledger Automation System                            |
| 5      | Hemasree Koganti & Srinivasa Anne | 2026 | Operational Excellence Through AI and ML in Financial Services: A Comprehensive Review of Applications, Challenges, and Future Directions          |



# Project Planning



## Module 1: User Interface Module

- Provides a menu-driven interface for users.
- Accepts user inputs and displays account information.
- Enables navigation through different banking operations.

## Module 3: Data Storage Module

- Stores account details using file handling.
- Maintains transaction history records.
- Retrieves saved data whenever required.

## Module 2: Banking Operations Module

- Creates and manages customer accounts.
- Performs deposit and withdrawal transactions.
- Calculates simple and compound interest.

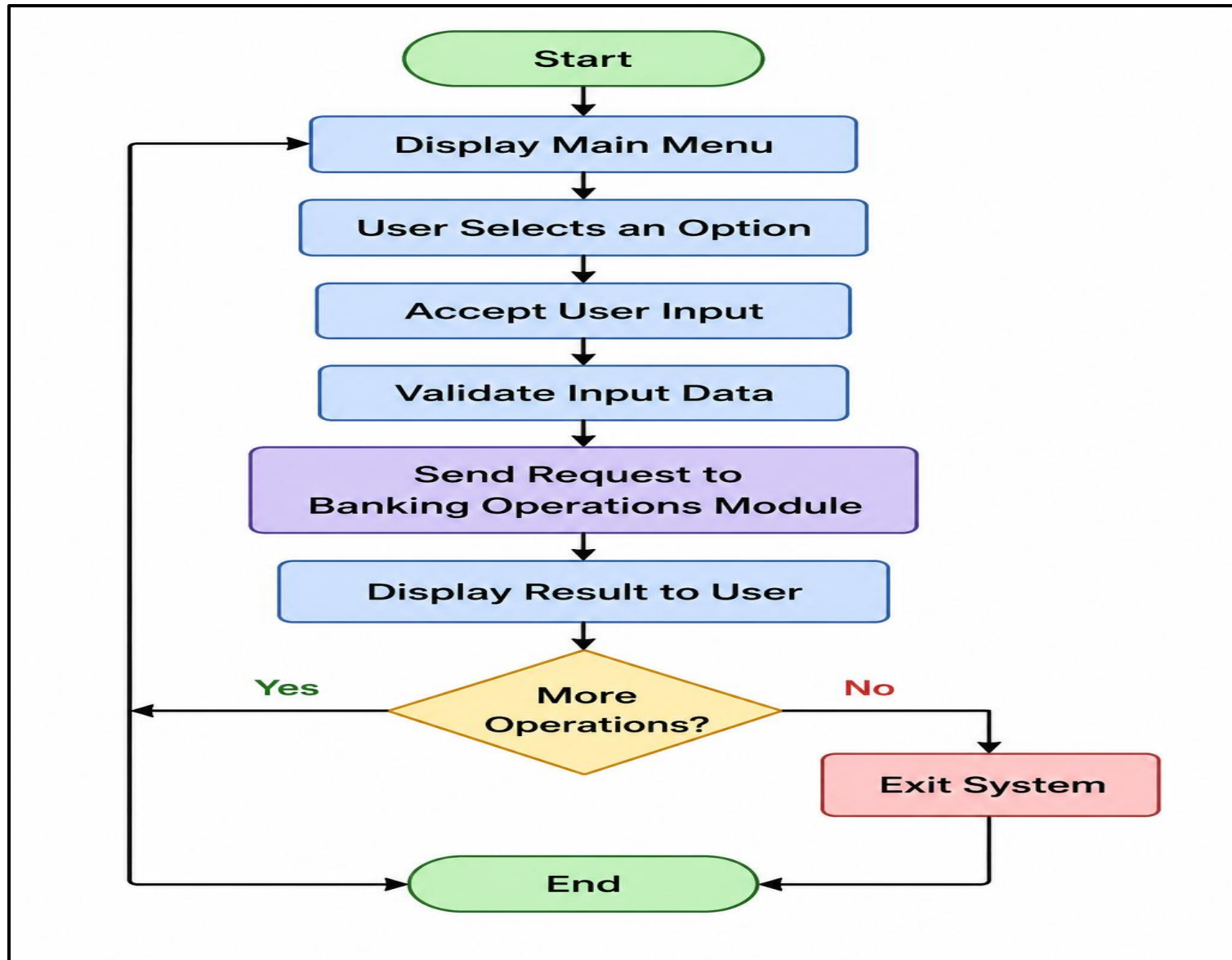
## Module 4: Account Security and Validation

- Authenticates users using Account ID and PIN.
- Validates user inputs to prevent invalid transactions.

## Module 5: Report Generation and Account Summary

Displays transaction history and current account balance.

# METHODOLOGY







# Module 1: User Interface Module



- Acts as the interaction layer between the user and the system.
- Displays options for account creation, transactions, and interest calculation.
- Collects account information and transaction details from the user.
- Processes menu selections and directs requests to the appropriate module.
- Displays account details, transaction history, and calculated interest results.
- Ensures smooth and easy navigation through the banking system.

The screenshot displays the VHA Federal Bank user interface. On the left, a sidebar lists five accounts for John Doe, Alice Smith, Venkatesh, Harsha, and Amarnath, each with a balance and account type (SAVINGS or CURRENT). The main area shows the 'ACTIVE ACCOUNT' #1001 for John Doe with a balance of ₹5,000.00. Below this, there are tabs for 'Transaction History', 'Perform Transactions', 'Accrue Interest', 'Interest Simulator', and 'Reports & Statements'. The 'Transaction History' tab is active, showing a 'Ledger Audit Logs' table with columns for DATE, TYPE, AMOUNT, RESULTING BALANCE, and DESCRIPTION. The table contains one entry: a deposit of \$5000.00 on 2026-01-01, resulting in a balance of \$5000.00, described as 'Initial Deposit'. At the bottom of the sidebar is a button to '+ Open New Account'.

| DATE       | TYPE    | AMOUNT     | RESULTING BALANCE | DESCRIPTION     |
|------------|---------|------------|-------------------|-----------------|
| 2026-01-01 | DEPOSIT | +\$5000.00 | \$5000.00         | Initial Deposit |



# Module 2: Banking Operations Module



- Receives user requests from the User Interface Module.
- Verifies account details before performing any operation.
- Creates new accounts and manages existing account information.
- Processes deposit and withdrawal transactions securely.
- Calculates simple and compound interest based on account data.
- Updates account balances and transaction records after each operation.
- Sends the processed results back to the User Interface Module.

## BANKING OPERATIONS MODULE

— Core Banking Operations and Processes —







# Module 3: Data Storage Module

- Receives account and transaction data from the Banking Operations Module.
- Stores customer account information in data files permanently.
- Records all deposit, withdrawal, and interest transactions.
- Retrieves saved account details whenever requested by the system.
- Updates existing records after each banking operation.
- Maintains transaction history for future reference and auditing.
- Ensures data persistence even after the application is closed.



# Module 4: Account Security and Validation

## Module

- Verifies user credentials for secure account access.
- Validates user inputs to prevent invalid or incorrect transactions.
- Ensures sufficient account balance before processing withdrawals.
- Protects customer data by restricting unauthorized access.



# Module 5: Report Generation and Account Summary Module

- Generates detailed account statements and transaction reports.
- Displays current account balance and interest earned.
- Provides a complete transaction history for each customer.
- Summarizes account activities to help users monitor their financial records.





## CONCLUSION



- Automates basic banking operations.
- Calculates simple and compound interest accurately.
- Maintains account and transaction records permanently.
- Provides secure deposit and withdrawal management.
- Offers a user-friendly menu-driven interface.
- Demonstrates practical use of C programming and file handling

**THANK YOU**